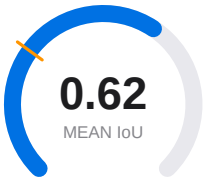
What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

FINDING

419622686 is the best physical fit.

Forward replay tested 4 candidates; the runner-up fits at 0.32. A ranking by physical consistency under tested assumptions — an investigative lead for a human analyst.



MEAN IOU	VARIANTS WON	LEAD RETAINED	CANDIDATES
0.62	**5** / 6	**96%**	**4**
the fit gate is 0.3	forcing variants where it fits best	of 80 tested changes	replayed by forward replay

EVIDENCE

- Best fit under **5 of 6** forcing variants
- Ahead of the runner-up by a mean loss gap of **0.1674**
- Inside the origin area **12 h** before the image
- The nearest vessel, **419838277**, 0.0035° away, is not supported

Candidates, ranked by physical fit

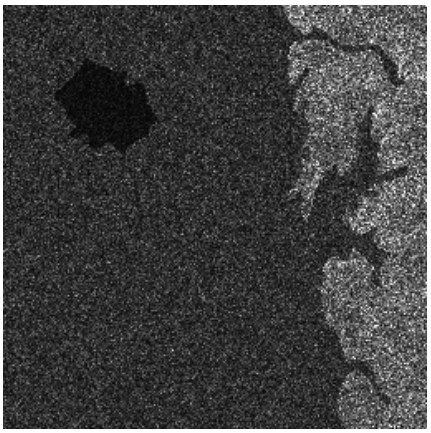
Each candidate's release replayed forward under six forcing variants and compared with the observed slick.

#	VESSEL	MEAN IoU		VARIANTS WON	RELEASE STARTS
1	419622686		0.625	5 of 6	9 Sep 2026, 23:30 UTC
2	419838277		0.323	1 of 6	10 Sep 2026, 03:00 UTC
3	419940288		0.331	0 of 6	10 Sep 2026, 02:30 UTC
4	419523658		0.040	0 of 6	9 Sep 2026, 11:30 UTC

The proximity baseline — the vessel nearest the slick when the image was taken — would pick **419838277**, ranked 2 by forward replay.

The scene

The radar image the slick was detected in. Radar darkness alone is not proof of oil.



Acquired	10 Sep 2026, 04:30 UTC
Sensor	synthetic SAR-like (VV proxy, single channel)
Pixel spacing	433.59 m
Data status	synthetic
Label source	generator ground truth (synthetic) — not a human or independently confirmed label
Projection	EPSG:32643

Robustness

Does the leader survive small changes to the model's assumptions?

96%
LEAD RETAINED

80 tested changes to the model's assumptions; **3** changed the leader.

The smallest change that flips it: **diffusivity multiplier 1.8** (magnitude 1.0), under which **419838277** leads.

Single-axis and two-axis perturbations only. Three-or-more-axis combinations were not tested, so absence of a flip is evidence of stability across the tested set, not a proof of stability.

Forward exposure

Where the oil is modelled to drift after the image. Reanalysis-style retrospective field — not an operational forecast.

AFTER THE IMAGE	MODELLED SPREAD	SHORELINE	FISHING GROUND	MANGROVE RESERVE
+6 h 10 Sep 2026, 10:30 UTC	497 km^2	0%	43%	0%
+12 h 10 Sep 2026, 16:30 UTC	543 km^2	0%	40%	0%
+24 h 11 Sep 2026, 04:30 UTC	590 km^2	1%	28%	0%

Traffic filtering

Every vessel in the scene, checked against where the oil could have come from over the 24 hours before the image.

1	Excluded	track data at every queried time, never inside the envelope
4	Candidate	track data present and inside the backward envelope at some queried time
1	Unassessed	never inside where track data exists, but no track data at one or more queried times — missing evidence, NOT exoneration

Unassessed is reported separately from excluded: missing evidence is not exoneration.

Coverage-aware matching

Matching radar contacts to AIS when some AIS is hidden — tested on synthetic contacts.

RATE	VESSELS HIDDEN	AREA GOES DARK
Hidden AIS recovered	**83%** 60/72	**84%** 91/108
False alerts	**26%** 32/124	**14%** 19/136
Mistaken promotions	**30%** 12/40	**59%** 20/34

Provenance

AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.
Generated	26 Sep 2026, 17:12 UTC
Software	Python 3.12.14 · Darwin 25.2.0 (arm64)

File integrity

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

attribution_result.json	`433bcb1f0af48a757e9acca5bac571f832f4d6f0c4789b08b28e784e2693b318`
coverage_result.json	`a9e8ec76d9439c11a160cabaa9d52154c363fee616ffd61f985b005e0e55c5fe`
forcing.npz	`f83ec6f9103d4c517a5c1e9f4196610e66c98e9c1b00e3d3cbb4471ebeb9f94e`
forecast_result.json	`626ecc1551295cef0484cdd85db97990f786943eae7134b13e0ff3160ab1c7a3`
fragility_result.json	`1d7d190a20f9512ea3f2668067f90809e3ce71604b669f68a093760d1e8fa92f`
historical_mapping.json	`cf29b589bba663c3c38263483116c3eb77acfd85e1ef4b2f92b81b95007b24a4`
manifest.json	`5ea96e0e5e1e7c371c2ec631ebc21721edf9b04c350046491b18f32ed0ca9e9e`
mask_groundtruth.png	`1627d69ebd773e78c268ea9fc333ebf3fe39b9f020c25dd638584bebf8d37568`
naive_baseline_result.json	`55b51daeeb5f875aedc4deb7c2ff2ad227be279121c3d5eb5dbd9106bf084296`
next_observation_result.json	`edf64bd0451e7b8c79bc79cdea82894fa45d0a5654f685c0197c707deb915e38`
scene.png	`065379b340b4e55ea310e48fac3c38c2859082312ad515844ee0783acb4f5012`

Uncertainty

— Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

— Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

— Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

— Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.